AIBEL JOMON

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EDUCATION

Texas A&M University (TAMU), College Station, Texas — August 2025 - May 2029
Bachelor of Science in Mechanical Engineering and Electrical Engineering Minor
Relevant Coursework: Engineering Computation, Statics & Dynamics, Materials & Manufacturing, Calculus II & III
Major GPA: 4.0, Cumulative GPA: 4.0

RELEVANT EXPERIENCE

SAE Baja - Controls Sub-Team Member, College Station, Texas — April 2026 - Current

Suspension DAQ & Controls Architecture

- Engineered CAN network and selected sensor suite for suspension data acquisition, maintaining bus load under 50% and node synchronization within 1 ms.
- Performed Hardware-in-Loop (HIL) and Software-in-Loop (SIL) testing to validate CAN architecture.
- Co-developed electronic hardware & enclosures and verified sensor accuracy for inertial dyno.
- Integrated sensor mounts and custom PCBs in SolidWorks, utilizing 3D printed PETG and adhering to IP67 ingress rating.

Tire Stiffness Rig

- Led end-to-end development of test rig to capture vertical tire stiffness data, improving quarter-car simulation accuracy and generating pressure-to-stiffness lookup tables.
- Performed FEA modeling and material selection in SolidWorks and Ansys to minimize beam deflection and guarantee measurement accuracy within ± 0.16 in.
- Designed pneumatic actuation assembly to load tire accurately, reducing force measurement error by 61.4%.

Naval Design Team - Electric Boat Sub-Team Member, College Station, Texas — August 2025 - Current

- Created custom 3D-printed mounting brackets for high-current speed controllers and transmitters for Promoting Electrical Propulsion (PEP) Competition boat in Fusion360, prioritizing Design for Manufacturing (DFM) principles.
- Configured transmitter for differential thrust, optimizing boat steering radius and implemented a kill switch.
- Configured receiver to send signals to throttle motors, achieving ~40 lbs thrust in water at 50% power output in tests.

PROJECTS & ORGANIZATIONS

Baja Design Challenge – Mechanical Braking Assembly, Personal Engineering Project — March 2026 - April 2026

- Designed SAE 2026 rules legal brake out of 6061-T6 Aluminum to handle 2,200+ lbf stomp force, exceeding 450 lbf requirements by 388%.
- Validated structural integrity through SolidWorks linear static FEA, engineering an I-beam profile that maintained Factor of Safety of 4.97 while only weighing 1.03 lbs.

American Society of Mechanical Engineers (ASME), Member — August 2025 - Current

- Attended monthly meetings, engaged with industry representatives to analyze current engineering challenges.

Remote Operated Vehicle, Personal Engineering Project — July 2025 - September 2025

- Developed V1 prototype using MicroPython on an ESP32 to execute PWM motor control via tethered 10ft UART communication.
- Architected V2 in C using the ESP-IDF framework, leveraging dual-core task allocation to isolate wireless UDP command processing from deterministic motor actuation.

TECHNICAL SKILLS

- **CAD & Design**: SolidWorks, FEA, ANSYS, Fusion 360, Design for Manufacturing (DFM).
- **Embedded Systems**: Microcontrollers (ESP32, STM32, Raspberry Pi, Arduino), Soldering, I2C, SPI, UART, ESP-IDF framework, CAN bus, Analog and Digital Sensors, Python, Embedded C, Rust, Linux.
- **Manufacturing**: 3D printing (FDM), Rapid Prototyping, CNC, Waterjet, Stick Welding Fundamentals.